Program: Data Science

Course Title: Computer Programming Foundations for Data Science (MSDS 510-01)
Semester and Year: *Fall* 2026

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2 Course Description:

Introduction to the basic foundations of computer programming for data science, using Python, R, or SAS as problem solving tools. 3 credit hours, Fall.

3 Course prerequisites:

None. Students are expected to have a working familiarity with the discipline of data science and analytics and general knowledge about the impacts of Big Data in businesses and corporations. All students should have a working knowledge of all aspects of Microsoft Office; and it goes without saying that they should be familiar with Internet access and usage.

4 Instructor:

- Name: M. Zakaria Kurdi
- Email Address: zakaria.kurdi@mmc.edu
- Building & Room: 3401 West End
- Day and Time of Class Meeting: Tuesday, 5:30-8:30pm
- Meeting Link: Blackboard Class
- Office Hours: Tuesday 4:30-5:30pm (Online Meeting with Appointment)

5 Course Objectives:

- Introduction to Python.
- Python syntax to write basic computer programs;
- Built-in and user-defined functions;
- Data Structures (arrays, lists, dictionary, tuple, etc)
- Introduction to object-oriented programming in Python.
- Handle File I/O
- Data Visualization

6 Learning Outcomes:

Students will be able to:

- Identify, compare, and explain the syntax and use cases for fundamental Python data types (string, list, integer, decimal, Boolean, dictionary), distinguish between mutable and immutable types, and describe how common operators, functions, and methods interact with these data types.
- Apply core programming constructs to develop interactive Python programs.
- Import and use modules for automation, file manipulation, and working with APIs; describe and process structured data from files (.txt, .csv); and transform, iterate through, and manipulate data from nested lists, dictionaries, and APIs for program needs.

- Assess program requirements to create and manage appropriate variables and data structures, design solutions for data analysis problems, and explain best practices for legally, ethically, and responsibly using genAI, including crafting effective prompts and understanding genAI outputs.

7 Class Format

Course content will be delivered through a combination of lectures, reading assignments, hands-on laboratory exercises, videos, podcasts, webinars, case studies, research-paper discussions, project presentations, and discussions with the instructor, guest speakers, and other students.
The course will follow a blended format. The instructor will make extensive use of **Blackboard** to distribute course materials, assignments, laboratory exercises, announcements, and other resources. From time to time, recorded lectures and supplementary instructional materials may also be posted on Blackboard.

8 Classroom Participation

Students are expected to have access to a computer capable of running the required software and to participate actively in hands-on exercises during class. Active participation is an important component of the learning experience. Students are expected to:

- Fully participate in classroom activities, laboratory exercises, discussions, research-paper discussions, and project-related activities.
- Keep their camera ON during online lectures and be dressed appropriately for an academic environment.
- Attend lectures from a location with a reliable network connection that allows full participation in classroom activities.
- Refrain from participating in other activities during lecture time, including driving, walking, attending another meeting, or engaging in activities that interfere with participation in the course.
- Join synchronous class sessions using a laptop or desktop computer rather than a mobile phone or other mobile device that limits the student's ability to participate fully in lectures, discussions, and laboratory exercises.
- Have the required course software installed and operational and be prepared to participate in exercises during class.
- Be prepared to discuss assigned readings, software implementations, and respond to questions related to assignments and project work.

Because this is a graduate-level course, students are expected to demonstrate professionalism, academic engagement, and preparedness throughout the semester.

9 AI Usage

Students are encouraged to use current technological tools, including Generative Artificial Intelligence (Generative AI), to enhance their understanding and review of course materials. AI tools may be used as learning aids to explain concepts, generate additional practice problems, explore alternative solutions, review code, and support independent study.
The use of AI tools, however, must support learning **rather than replace the student's own understanding and work**. The following requirements apply:

- AI should not replace students' own reasoning, analysis, or authorship.
- Use of AI in graded assignments or other evaluated work should follow the instructor's specific permission or restrictions.

- Do not upload confidential, proprietary, personally identifiable, patient-identifying, credential, or otherwise restricted data to public Generative AI tools.
- Students may use AI tools to review course concepts, practice skills, and solve additional problems beyond assigned coursework.
- For assignments, projects, and other submitted work where AI use is permitted, students must disclose the use of AI tools and clearly identify the sections, code, analyses, or other components for which AI assistance was used.
- Students remain fully responsible for the accuracy, correctness, originality, and completeness of all submitted work, including content generated or suggested by AI.
- Students must be able to explain and defend all submitted work regardless of whether AI tools were used.
- During quizzes, examinations, designated in-class exercises, or other activities identified by the instructor, the use of AI tools may be prohibited. Students must follow the specific instructions provided for each assessment.
- Students are expected to develop sufficient proficiency to solve course problems independently without assistance from AI tools when required.
- Undisclosed or unauthorized use of AI tools may be treated as an academic integrity violation in accordance with institutional policies.

10 Grading Scheme

Students will work on various lab (programming) assignments, outside of class, which will be weighted as follows:

Grading Element	**% Weight**
Attendance and Class Participation	05
Assignments	20
Reading Assignments & Discussion Questions	05
Quizzes & In-class Exercises	20
Midterm Exam	10
Final Exam	20
Project	20

Grades will be posted and maintained on Blackboard. Whenever possible, graded work will also be returned through Blackboard. While some materials may be distributed during class sessions, all course-related materials (assignments, schedules, and announcements) will always be available on Blackboard. To the extent possible, all submissions must be completed electronically via Blackboard or other approved platforms.

10.1 Code Review

For both assignments and the course project, students should be prepared to present and explain their code during class sessions. This includes answering questions related to design decisions, implementation, and functionality.

10.2 Late Submission

- Late submissions are generally not accepted. All work must be completed by the specified deadlines.
- There will be 50% penalty for assignments late by one week.
- There will be NO CREDIT (0 marks) for the work late by more than one week.
- In the case of a documented emergency, students must contact the instructor as soon as possible.

10.3 Grading Timeline

Please be aware that grading of written assignments and homework takes time, in the same way that it takes time for students to prepare them. The instructor's goal is to promptly give feedback on all assignments; but there may be some assignments that require considerable time to grade. It is important to make sure that everything each student prepares and submits is thoughtfully and carefully considered.

10.4 Grading System:

Weighted averages will be computed at the end of the course. The grading system for the course is given in the following table.

Letter Grade	GPA	Semester Combined Average
A	4.00	90 or above
B+	3.50	85 – 89.9
B	3.00	80 – 84.9
C+	2.50	70 – 79.9
C	2.00	65 – 69.9
F	0.00	Below 65

There is no pre-established quota of specific grades for this class. All final grades shall remain on the student's permanent transcript.

10.5 Incomplete Grade

The grade of "I" (Incomplete) indicates that the student has satisfactorily completed at least three-fourths of a course, and may be given at the instructor's discretion to a student when illness, necessary absence, or other reasons beyond the control of the student prevent completion of course requirements by the end of the academic term. A student cannot receive an "I" if he/she is failing a course. A student receiving an "I" must complete the requirements for the course to remove the "I" by the end of the next semester the course is offered. If the requirements are not completed within the specified time, no credit will be given and the Office of Admissions and Records will automatically record the final grade as "F".

10.6 Withdrew Voluntarily (WV) and Withdrawn Administratively (WA)

The symbols "WV" and "WA" indicate that the student "Withdrew Voluntarily" or was "Withdrawn Administratively by the Dean." The symbols "WP" and "WF" indicate that the student "Withdrew Passing" or "Withdrew Failing," respectively. These symbols are used only when the student has withdrawn after at least six weeks of attendance in a course during the fall or spring semester or after two weeks during the summer. Whether 'WP' or 'WF', if the student returns to a particular course, he or she must take it over in its' entirety.

10.7 In-Progress Grade

The "IP" (In Progress) is awarded for certain courses that are continuous over more than one semester and, as such, are not finally evaluated until the conclusion of the sequence. Final grades are given in such courses only at the end of the final semester of the course sequence. A record of academic progression, however, shall be reported in the Office of Admission and Records at the end of any given semester using the designation In-Progress (IP). Quality points will be calculated using the total hours of the course.

10.8 Grades for External Students

Students from other programs will receive a letter grade for the course per the requirement of their individual programs. All students will be expected to attend all class sessions, complete all assigned readings and participate in all group assignments.

11 Expectations of Students:

11.1 Brand Mantra

The three words that articulate the "heart and soul" of the SACS are Innovate, Connect, and Serve. These three words stand out in our Mission Statement.

11.2 Overall Goal (Vision)

To be recognized for our impact on society.

11.3 Overall Purpose (Mission)

The School of Applied Computational Sciences advances scholarship in key domains of computational sciences by Innovating through academic excellence and impactful research, connecting to key health care industries, businesses, influencers, and thought leaders in the areas of computational science, and Serving society at large.

11.4 Student Expectations

SACS expectations for student learning align with its mission statement. SACS is here to assist its students in reaching their academic, professional, and personal goals, and to instill hearts of service and compassion in them. As a SACS student, you will be expected to:

- Refer to your syllabus throughout the semester;
- Learn and think critically, and take time to ask questions when you do not understand something;

- Build problem-solving skills by attempting to solve the problem(s) yourself first;
- Maintain relationships with your professors (and advisors) and report to them any degree of difficulty that you may experience with your courses;
- Come to every advising meeting prepared with questions, materials and/or topics to be discussed;
- Accept full responsibility for your academic success and acknowledge that academic advisors are only one of many resources for achieving success;
- Be a self-advocate and learn how to locate and utilize available institutional resources and student support services;
- Embrace service, professional development, and research activities to prepare for your future career opportunities;
- Make connections between classroom instruction and the real-world and get involved beyond the classroom;
- Demonstrate respectful, courteous, and professional behavior, all-round, and when communicating with faculty in person or through e-mail; and
- Display an attitude toward work that includes integrity, responsibility, and a desire for excellence.

12 Faculty Expectations

SACS faculty will create a learning environment in which people treat each other with dignity, courtesy and respect, while at the same time promoting an open and critical exchange of ideas. To do this, SACS faculty will be expected to:

- be knowledgeable about the subject matter;
- be prepared for class with materials and visuals designed with all students in mind, including those with disabilities;
- approach each class with enthusiasm for the learning process and academic inquiry;
- foster an environment of collaborative learning, encouraging active participation rather than passive attendance;
- respect students as individuals, considering each student's learning style, background, and demands on time;
- set high performance standards and communicate how to achieve them successfully;
- clearly state standards for evaluating student work when assignments are given;
- return student work or respond to student inquiries in a timely fashion;
- honor office hours and be reasonably available at other times; and,
- refrain from using language, humor, or course examples that may be insulting or insensitive.

From time to time, course participants may communicate with the instructor and classmates via email or social media. Email, in particular, tends to be a more awkward form of communication. Things said out loud to someone may seem impolite or even inflammatory in an email situation. THINK BEFORE HITTING SEND! Always reread the message and ask "could this offend someone?" Your instructor is committed to developing and actively promoting a class environment in which respect is shown to everyone in order to facilitate and encourage the expression, testing, understanding, and creation of a variety of ideas and opinions. Rude, sarcastic, obscene, or disrespectful messages/comments/remarks have a negative impact on everyone's learning and will not be tolerated. Students are expected to abide by the University's policies and principles of academic integrity.

Students enrolled in this course are expected to conduct their affairs ethically with honesty and integrity. Individuals who do otherwise can expect their professional careers to be adversely impacted. Unless otherwise indicated, all assignments are to be completed individually, with help from no one other than the instructor when warranted. It goes without saying that group assignments require collaboration among students. You should check with your instructor if you have concerns about group work.

Unethical academic behavior (cheating) includes, but is not limited to, copying assignments or answers on exams, plagiarism, inappropriate or fraudulent citation of the work of others, "borrowing" other students' data/information and submitting/representing it as your own, unauthorized copying of computer files, and unauthorized use of electronic media to transmit information and communicate with other students at test time (email, instant messaging, use of chat rooms, etc.). At a minimum, students who cheat will receive "no credit" (a score of zero) on the assignment in question; but students may also be dismissed from the course and automatically assigned a grade of F. In addition, any suspected evidence of cheating will be referred to the Department Chair.

The article "Integrity: Academic and Political—A Letter to My Students," by Bill Taylor is an excellent discourse concerning academic integrity and each student is encouraged to read it.

https://integrityseminar.org/wp-content/uploads/2018/02/Taylor-Letter-to-My-Students.pdf

13 Course Withdraw Policies

Any student who no longer attends class after course registration and drops the course either voluntarily or due to some reasons is required to inform the instructor first and then submit the request in Workday to drop the course. Meharry has college wide policy regarding the date of course drop/withdraw that may affect the grade and finance.

14 Required Textbook:

- **Python for Everybody** by Chuck Severance. Available on Python for Everybody

15 Class Topics

The following is a **tentative** schedule.

Date	Lecture #	Week #	Material Coverage
Aug 13	1	Week 1	Syllabus & Python environment setup;
Aug 20	2	Week 2	Variables and datatypes; Arithmetic Operators
Aug 27	3	Week 3	Conditional Statements (if else) – Logical Operators
Sep 3	4	Week 4	Iterative Statements (for, while)
Sep 10	5	Week 5	Functions
Sep 17	6	Week 6	Data Structures (lists)
Sep 24	7	Week 7	Data Structures (dictionary and tuples)
Oct 1	8	Week 8	Files (Input, Output, Exception Handling, datasets)
Oct 8	9	Week 9	Classes and Objects (Object Oriented Programming)
Oct 15	10	Week 10	Classes and Objects (Object Oriented Programming)
Oct 22	11	Week 11	Numpy (matrices)
Oct 29	12	Week 12	Pandas (Exploratory Data Analysis)
Nov 5	13	Week 13	Data Visualization (charts, plots, heatmaps)
Nov 12	14	Week 14	Project Presentations
Nov 19	15	Week 15	Project Presentations

16 Important Academic Dates

Date	University Event	Course Impact
Mon., Aug. 10	Fall 2026 Classes Begin	MSDS 510-01 begins Wed., Aug. 12
Mon., Aug. 24	Add/Drop Advising Ends	Reminder during Week 2
Mon., Sep. 7	Labor Day Holiday	Tuesday class (Sep. 8) meets normally
Fri., Sep. 25	Fall Census Date	Administrative reminder
Mon., Oct. 5	Convocation	Wednesday class (Oct. 7) meets normally
Mon., Nov. 9	Spring Advising Begins	Mention during Week 14
Thu., Nov. 19	Fall 2026 Classes End	All instructions completed by Nov. 17
Fri., Nov. 20	Grades Due in Workday	Final grades submitted

17. Miscellaneous Information:

- Instructor Availability and Response Time: Please feel free to email your instructor directly with questions of a personal nature, questions about grades, requests for help, or other issues that are not appropriate for the rest of the class to hear/read. Email is checked regularly, and responses are sent as quickly as possible. There may not be any responses on Saturdays, but every attempt will be made to check on Sunday evenings to make sure there are no pressing issues involving assignments that are due early the following week.
- Recommended Reading: As data science professionals, you should strive to keep abreast of the latest information about, and thought leadership in, the discipline.
- Class Interaction and Communication: Ongoing interaction with classmates and the instructor is vital to the success of the learning environment. Participation of this nature is required of all students. Some of the class interactions may entail postings to the discussion board on the Blackboard. The discussion board will be checked on a regular basis and questions will be

answered as needed. Students are strongly encouraged to read and respond to postings from their classmates to facilitate problem troubleshooting and the development of critical thinking skills. Responses should be substantive and thought-provoking, referring to specific material that is related to the course. In most cases, responses such as "*I agree*" are insufficient. All class members may periodically meet with the instructor at a virtual conference (e.g., via Zoom).

About Instructor:

https://meharry.edu/faculty_staff/kurdi-zakaria/

Dr. Mohamed Zakaria Kurdi is an Associate Professor at Meharry Medical College in Nashville, Tennessee, specializing in applied natural language processing (NLP) and text mining for healthcare, particularly Alzheimer's disease (AD). His research focuses on developing speech- and text-mining pipelines alongside state-of-the-art AD diagnostic classifiers. He is also the author of a seminal two-volume introductory Computational Linguistics textbook published in English and French.

Dr. Kurdi holds a Ph.D. in Computer Science from Université Joseph Fourier, France. His extensive academic career includes teaching at the University of Lynchburg (2016–2025), serving as a visiting professor at the University of Delaware, and acting as a visiting researcher at the University of Ottawa. Additionally, he spent over two years as a postdoctoral researcher at the University of Southern Denmark working on European dialog systems and NLP projects.

Recognized with accolades such as the 2022 MLDM Best Paper Award, his future vision involves pioneering multimodal NLP for real-time AD trial matching and precision interventions.